

Evaluating the Performance of LSTM and GRU in Detection of Distributed Denial of Service Attacks Using CICDDoS2019 Dataset



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Abstract A Distributed Denial of Service (DDoS) attack occurs when an intruder or a group of attackers attempts to block legitimate users from accessing a service. A DoS attack is carried out by a single system, while a DDoS attack is carried out by numerous systems. DDoS attacks can be directed at several OSI layers. Deep learning has played a crucial role in the advancement of intrusion detection technologies in recent years. The main purpose of this work is to detect and identify DDoS attacks in the OSI model's application, network, and transport layers using deep learning models. The proposed models have been evaluated against the CICDDoS2019 dataset which consists of application, network and transport layer DDoS attacks. For the CICDDoS2019 dataset, Long-Short-Term memory and Gated Recurrent Unit attained an average accuracy of 99.4% and 92.5%, respectively. We also compared the suggested models' performance to that of a few other higher accuracy models and found that the proposed models have higher accuracy with fewer epochs. In addition, the performance of the proposed system is also evaluated for various types of DDoS attacks in the CICDDoS2019 dataset and LSTM is found to produce good accuracy.

Keywords Distributed Denial of Service (DDoS) · Deep neural network · Long short-term memory · Gated recurrent unit · CICDDoS2019 · Accuracy

1 Introduction

An Artificial Neural Network (ANN) is a computer system that is capable of learning and making intelligent and prudent decisions through the use of algorithms. Deep learning is a subtype of neural network. A neural network with multiple hidden layers of nodes is used in a deep learning model. The term “deep” refers to the number of layers in the neural network. Deep learning was developed primarily for the purpose of managing massive amounts of data and running complex algorithms

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J. H. Kim et al. (eds.), *Proceedings of 7th International Conference on Harmony Search, Soft Computing and Applications*, Lecture Notes on Data Engineering and Communications Technologies 140, https://doi.org/10.1007/978-981-19-2948-9_38

395

to optimize performance. Deep learning models are used for both feature extraction and classification. Data flows from the input layer to the output layer of a Deep Neural Network (DNN), and no loops or cycles are produced. A Recurrent Neural Network (RNN) is a type of ANN in which the connections between the nodes form a directed graph and can learn sequential data over time steps. When the back-propagation algorithm iterates across all the neurons in the neural network to update their weights, the vanishing gradient problem occurs. Long Short-Term Memory (LSTM) has been designed to address the problem of vanishing gradients in RNNs. The GRU is a more recent type of RNN that resembles an LSTM. The GRUs obliterated the cell state in favor of using the hidden state to transfer data.

Distributed Denial of Service (DDoS) is a malicious attempt to degrade or interrupt a server's legitimate traffic. DDoS attacks are classified into three types: volume-based attacks, transport/network-layer attacks, and application layer attacks. By flooding a resource, such as a server, with fake traffic, volume-based attacks overload it. ICMP, UDP, and spoofed-packet flood attacks are such attacks. DDoS attacks at the protocol levels such as SYN floods and Smurf DDoS, flood a targeted server with packets. Application layer attacks are initiated by overflowing apps with bogusly created requests. DDoS attacks may also appear to be benign traffic. Even a big volume of legitimate requests from legitimate users will bring the system to a grinding halt. As a result, research into the detection of DDoS attacks is becoming increasingly critical. Machine learning is a widely used technology for detecting DDoS attacks through the use of statistical features. On the other side, machine learning methods employ shallow representation models, which do not result in increased accuracy. However, deep learning models have recently proved their ability to discriminate between DDoS and legitimate traffic by discriminating between high-level and low-level properties.

Additionally, we noticed that the bulk of DDoS attack detection techniques evaluate performance using the NSL-KDD and KDDCUP 99 datasets. Since these datasets have been created long back and do not reflect the recent trends in the attack pattern, we use CICDDoS2019. We create two models using LSTM, and GRU to train and recognize the types of attacks in this dataset. DDoS attacks can be classified as network/transport layer or application layer attacks, depending on which tier the DDoS attacker targets. The CICDDoS2019 dataset comprises traces of attacks on the application, transport, and network layers. Additionally, we employ the proposed approaches to detect other sorts of attacks in the dataset, including LDAP, UDP, Portmap, NETBIOS, and UDP-LAG. Classification of harmful packets and neural network training serve as the foundation for deep learning-based attack diagnostics. The remaining part of the paper is set out as follows: Sect. 2 examines state-of-the-art deep learning models developed to detect DDoS attacks using contemporary and CICDDoS2019 datasets. We provide a brief summary of the dataset in Sect. 3. Section 4 describes how LSTM and GRU are used to classify the attacks in the dataset. In Sect. 5, we analyze the performance of the proposed models and present our results. Finally, we provide our conclusions in Sect. 6.

2 Related Works

Using big data and neural networks, Hsieh and Chan et al. [1] suggested a DDoS detection technique. The detection system was written in R and is based on Apache Spark. Ferrag et al. investigated a variety of deep learning models for detecting cyber security intrusions and reviewed the two new real-time traffic data sets CSECI-CIDS2018 and the Bot-IoT [2]. Corin et al. proposed a system called “LUCID” that classified traffic flows as malicious or benign based on CNN features [3]. Asad et al. developed a framework for detecting DDoS attacks using a deep learning method. For CICIDS2017 dataset, this system employed a feed-forward back-propagation network to identify application layer DDoS attacks [4]. Anthi et al. developed an IoT IDS using machine learning algorithms. The system can differentiate network scanning, probing, and simple DDoS attacks [5]. Doshi et al. demonstrated a packet-level DoS system capable of consistently distinguishing between normal and DoS attack data received from IoT devices [6]. They demonstrated that by leveraging IoT-specific network behaviors and feature selection, it is possible to accurately detect DDoS attacks in IoT network data using a variety of machine learning algorithms. Elsayed et al. developed a DDoSNET, an IDS for detecting DDoS assaults in a Software Defined Network, using the CICDDoS2019 data set. This approach made use of RNN and Auto Encoder [7]. Jiang et al. created a system for anomaly identification using back-propagation neural networks. This framework incorporated traffic and user activity data extracted from web server logs. The authors assessed the performance of the proposed model using the CICIDS2017 data set [8].

Shone et al. (2018) proposed the Non-symmetric Deep Auto Encoder as an unsupervised feature learning approach. Classification of this model, which has been generated using stacked NDAEs, was performed using the Random Forest algorithm [9]. Hsieh and Chan [10] suggested two techniques for detecting Distributed Reflection DoS (DrDoS) attacks on the Internet of Things. In the first way, they employed a hybrid Intrusion Detection System (IDS) to detect IoT-DoS attacks, and in the second method, they trained a deep learning model based on LSTM on the most current data set to detect various forms of DrDoS attacks. Muraleedharan and Janet [11] describes a technique for identifying HTTP-related DoS threats using a deep classification technique based on traffic statistics. The classifier is evaluated using CICIDS2017 data. An attempt by Cil et al. [12] proposed employing a DNN as a deep learning model for detecting DDoS attacks on a batch of packets obtained from network traffic. The attack types were classified in this study with an accuracy rate of 94.57%. Bolodurina et al. [13] examined the effect of data balancing approaches in the network traffic categorization issue on various types of DDoS using the CICDDoS2019 dataset, which contains data on reflection- and exploitation-based attacks and achieved a maximum accuracy of 98.62%. Sindian and Sindian [14] suggested an improved Deep Sparse Autoencoder-based approach for identifying DDoS attacks that utilizes two hidden layers. The primary objective of this work is to employ an autoencoder to extract representative features from the CICDDoS2019 dataset, to minimize classification error, and to correctly detect DDoS attacks. This

investigation achieved a detection accuracy of 98%. Koliass et al. [18] presented a DDoS attack on the IoT using Mirai and botnets. This attack was carried out via a network of interconnected devices connected to the Internet and a specific malware called “Mirai botnet,” which pummeling the targeted servers with traffic until they succumbed to the strain. Hou et al. [19] proposed a method for identifying DDoS traffic using NetFlow feature selection and machine learning. Additionally, they evaluated the CICIDS2017 dataset in conjunction with real-world NetFlow logs given by a large ISP in China, Unicorn. The experiment is conducted utilizing a random forest as the detector, and the findings obtained using the CICIDS2017 demonstrate 99% accuracy. Ferrag et al. [20] suggested deep learning model for cyber security intrusion detection and used two new real-world traffic datasets: CSECICIDS2018 and Bot-IoT. From the reviews, we believe that deep learning has potential to detect recent types of DDoS attacks and can be well explored to improve the detection accuracy. The different researches are used for detecting the application, network and transport layer-based DDoS attacks. Nevertheless, we could find only a few research works on CICDDoS2019 dataset. Moreover, these research attempts have found that whether a particular instance is an attack or not, but not the type of the attacks. But we find the different types of attacks also.

3 Dataset Description

Given the nature of DoS/DDoS attacks, this section provides a brief overview of benchmark datasets for designing systems for detecting DoS/DDoS attacks. DARPA, KDD, and NSL-KDD are just a few publicly available datasets that have been utilized as benchmarks. In this study, the dataset CICDDoS2019 [16] has been used to develop and assess deep learning models. This dataset includes the most recent and benign prevalent DDoS attacks on the application and transport/network layers. The CICDDoS2019 data collection includes 12,794,627 traces of DDoS attacks. Among the 86 features are flow length, total forward packets, total reverse packets, and so on [16]. The dataset is classified as benign, NETBIOS, UDP, UDP-LAG, LDAP, NTP, SSDP, TFTP, and MSSQL.

4 Deep Learning Models for DDoS Attack Detection

The primary objective of this study is to analyze the CICDDoS2019 data sets using deep learning. We employ three deep learning models in this study to detect DDoS attacks at the application, network, and transport layers. These models include a deep neural network generated using the feed-forward back-propagation approach and a RNN constructed using LSTM and GRU.

4.1 Preprocessing

Preprocessing is a technique for preparing data for analysis. When data is in its “raw” state, it is of limited utility for analysis. As a result, we perform preprocessing on the data before passing it to a classification system. To avoid overfitting, categorical features such as the source IP address, the destination IP address, the port number, and the source port are excluded. We have removed instances with attributes containing NAN and infinite values because we have a sufficient number of records. Due to the fact that all features should be within the same range, we can scale them using a variety of scaling strategies, including standard scalar and Minmax Scalar. Minmax Scalar has been utilized in our study to convert all of the features to a same set. Then, using label encoding, we substitute a numeric value between 0 and the number of class labels for the category values of the class labels. CICDDOS2019 has a class designation of 0 (for benign traffic) and 1 (for attack traffic). The term “attack traffic” refers to all forms of attacks. The next sections describe the deep learning models used to categorize the chosen dataset in this work.

4.2 Recurrent Neural Network Using LSTM and GRU

In a typical neural network, each neuron’s output is determined by the current input, with no relationship between the input and the neuron’s previous output. However, if we want to predict the next word in a sentence, we must recall the previous words to do so correctly. Thus, RNN was introduced. Unlike feed-forward neural networks, RNN have cyclic connections thus making them more appropriate for modeling sequence of data. As a result, in recent years, RNN has played an important role in machine translation, robot control, time series prediction, and other areas. Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) are two types of RNN.

LSTM: When training, conventional RNN faces the issue of vanishing gradients. The LSTM algorithm is used to solve this problem. RNN cannot carry information from earlier time steps to later ones for long-term dependencies if a series is long enough. Hence, LSTM is a good choice for sequences with long-term dependencies.

GRU: An LSTM without an output gate is known as a GRU. At each time step, GRU is able to write the contents of memory cells to the larger net. The vanishing gradient problem in RNN is also addressed by GRU. Since both are built similarly and, in some cases, produce equally good results. RNN and its variants are shown in Figs. 1 and 2.

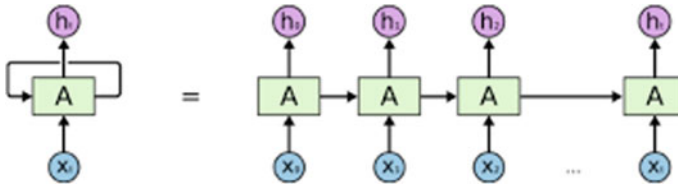


Fig. 1 A simple RNN

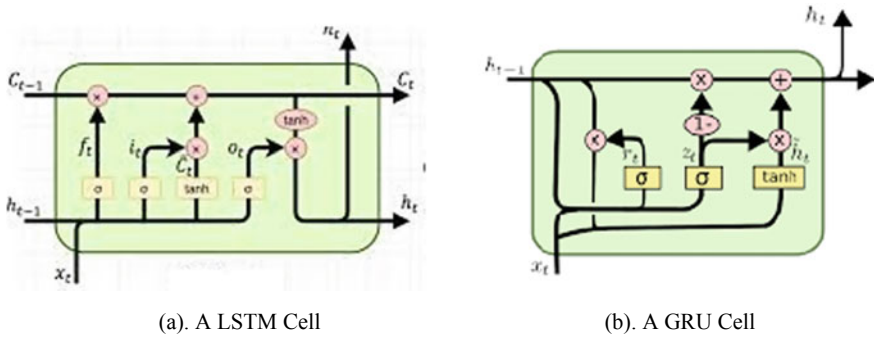


Fig. 2 RNN and its variants

5 Results and Discussion

This section presents and discusses the outcomes of the experiments conducted using the neural network models that have been developed. DELL EMC 740 with 128 GB RAM and 32 Gb GPU RAM is utilized to train models in this article. Hyper-parameters are critical in deep learning algorithms because they describe the details of training and have a direct impact on model output. We used Bayesian optimization to select optimal values for hyper-parameters that would yield good accuracy. It is a search method that employs Bayes' theorem to find the minimum or maximum of an objective function. This study uses accuracy as the objective function, which we would like to maximize. The Bayesian approach may increase the time spent on selecting the hyper-parameters but reduce the time spent evaluating the objective function, resulting in a lower computational cost. The set of hyper-parameters tuned in this study is optimizer, learning rate, activation function, and number of neurons. Other parameters like batch size, epochs, etc. are fixed. Table 1 summarizes the hyper-parameters tuned in our work and their search spaces.

A summary of the LSTM and GRU training descriptions can be seen in Table 2.

For the first set of experiments, we have translated the eight labels of the attack class into “attacks” in the CICDDoS2019 dataset. So, in the output layer, we used binary cross-entropy as a loss function. The data set CICDDoS2019 is divided into training and test sets in an 80/20 ratio. Training data is given for the proposed neural

Table 1 Hyper-parameters and their search space

Parameter	Search space	Description
Optimizer	Adam, RMSProp, SGD	To optimize the input weights by comparing the prediction and the loss function
Learning rate	1e−3, 1e−4, 1e−5, 1e−6	To determine the step size at each iteration while minimizing the loss function
Activation function	Relu, Elu and Tanh	To introduce non-linearity into the output of neurons
Number of neurons	64,128, 56, 512, 1024	To compute the weighted average of input

Table 2 Tuned hyper-parameters in LSTM and GRU

Parameters	Values
Optimizer	Adam
Loss	Binary cross-entropy/categorical cross-entropy
Activation	Relu
Learning rate	0.001
The number of neurons	1024

network models. Their performance against DDoS attacks for test data sets were assessed after training of LSTM and GRU. We used the hyper-parameter settings in Table 2 to run the built models. The models were run with a varied number of layers, and the results are shown in Table 3. From Table 3, it is understood that LSTM provides better classification accuracy for CICDDoS2019 data set. Due to the long-term dependent nature of LSTM, it produces good accuracy. In second set of experiments, we have evaluated the detection performance of the proposed models against each type of the attacks. Various metrics such as Accuracy, Precision, Recall, and F1-score were evaluated and presented in Tables 4, 5, 6, 7, 8, 9, 10 and 11.

As can be seen from Tables 4, 5, 6, 7, 8, 9, 10 and 11, LSTM outclassed GRU for all the types of attacks. Because of the long-term dependencies of the dataset, LSTM performed well. Another reason for better accuracy by LSTM is that the size of the

Table 3 Performance evaluation of LSTM and GRU on CICDDoS2019 dataset

Metrics	Epochs = 20					
	LSTM			GRU		
Number of layers	1	2	3	1	2	3
Training accuracy (%)	99.1	99.32	99.41	92.25	92.51	92.73
Testing accuracy (%)	99.02	99.43	99.38	92.87	93.07	93.55
Precision (%)	98.8	99.04	98.83	93.8	93.9	94.1
Recall (%)	98.56	98.3	99.12	92	92.67	92.9
F1-score (%)	99.01	99.3	99.23	92.9	93.29	93.56

Table 4 Performance evaluation of LSTM and GRU against LDAP

No. of layers	No. of epochs	Training accuracy (%)		Testing accuracy (%)		Precision (%)		Recall (%)		F1-score (%)	
		LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU
1	5	90.07	90.84	90.67	91.20	84.41	84.4	99.21	99	91.56	91.5
1	10	90.80	90.96	91.05	91.35	84.46	84.4	99.43	99	91.55	91.5
2	5	90.55	90.14	90.71	90.78	84.42	84.4	99.12	99	91.50	91.5
2	10	92.62	90.50	93.13	91.34	84.44	84.4	99.37	99	91.52	91.5

Table 5 Performance evaluation of LSTM and GRU against NETBIOS

No. of layers	No. of epochs	Training accuracy (%)		Testing accuracy (%)		Precision (%)		Recall (%)		F1-score (%)	
		LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU
1	5	95.87	94.19	96.09	94.94	92.9	92.9	99.9	99.9	96.3	96.35
1	10	96.29	94.75	99.15	95.0	92.9	92.9	99.9	99.9	96.3	96.3
2	5	95.13	94.45	94.85	99.30	92.9	92.9	99.9	99.9	96.3	96.3
2	10	94.02	95.03	94.99	94.74	92.9	92.9	99.9	99.95	96.3	96.3

Table 6 Performance evaluation of LSTM and GRU against NTP

No. of layers	No. of epochs	Training accuracy (%)		Testing accuracy (%)		Precision (%)		Recall (%)		F1-score (%)	
		LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU
1	5	96.15	95.79	96.50	96.58	99.8	99.8	93.2	93.2	96.4	96.4
1	10	96.50	95.34	95.63	96.18	99.8	99.8	93.2	93.2	96.4	96.4
2	5	95.89	96.18	96.52	96.66	99.8	99.8	93.2	93.2	96.4	96.4
2	10	96.24	96.11	96.62	96.31	99.8	99.8	93.2	93.2	96.4	96.4

Table 7 Performance evaluation of LSTM and GRU against SSDP

No. of layers	No. of epochs	Training accuracy (%)		Testing accuracy (%)		Precision (%)		Recall (%)		F1-score (%)	
		LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU
1	5	92.71	92.32	94.16	93.71	89	89	99	99	94.1	94.1
1	10	94.42	90.35	95.32	92.86	89	89	99	99	94.1	94.1
2	5	91.24	94.35	95.44	95.37	89	89	99	99	94.1	94.1
2	10	93.84	93.04	96.87	95.80	89	89	99	99	94.1	94.1

Table 8 Performance Evaluation of LSTM and GRU against MSSQ L

No. of layers	No. of epochs	Training accuracy (%)		Testing accuracy (%)		Precision (%)		Recall (%)		F1-score (%)	
		LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU
1	5	93.18	92.81	94.11	93.01	88.1	88.1	99.9	99.9	93.6	93.6
1	10	92.87	92.99	93.04	94.31	88.1	88.1	99.9	99.9	93.6	93.6
2	5	92.06	92.42	92.93	93.04	88.1	88.1	99.9	99.9	93.6	93.6
2	10	92.92	92.94	93.05	93.05	88.1	88.1	99.9	99.9	93.6	93.6

Table 9 Performance evaluation of LSTM and GRU against UDP-LAG

No. of layers	No. of epochs	Training accuracy (%)		Testing accuracy (%)		Precision (%)		Recall (%)		F1-score (%)	
		LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU
1	5	93.25	88.78	93.77	88.07	90.7	90.7	98	98	94.2	94.2
1	10	93.25	91.46	94	92.75	90.7	90.7	98	98	94.2	94.2
2	5	90.32	88.75	90.74	93.31	90.7	90.7	98	98	94.2	94.2
2	10	91.80	91.49	94.01	92.98	90.7	90.7	98	98	94.2	94.2

Table 10 Performance evaluation of LSTM and GRU against UDP

No. of layers	No. of epochs	Training accuracy (%)		Testing accuracy (%)		Precision (%)		Recall (%)		F1-score (%)	
		LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU
1	5	92.9	91.57	93.72	93.33	94.6	94.6	99.9	99.9	97.2	97.2
1	10	95.67	93.91	95.73	94.78	94.6	94.6	99.9	99.9	97.2	97.2
2	5	92.56	91.39	93.97	95.45	94.6	94.6	99.9	99.9	97.2	97.2
2	10	94.69	94.56	96.19	95.52	94.6	94.6	99.9	99.9	97.2	97.2

Table 11 Performance evaluation of LSTM and GRU against TFTP

No. of layers	No. of epochs	Training accuracy (%)		Testing accuracy (%)		Precision (%)		Recall (%)		F1-score (%)	
		LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU	LSTM	GRU
1	5	87.75	87.66	89.83	90.41	83.4	83.4	99.4	99.4	90.7	90.7
1	10	88.85	90.27	91.72	91.07	83.4	83.4	99.4	99.4	90.7	90.7
2	5	89.28	90.40	91.16	91.46	83.4	83.4	99.4	99.4	90.7	90.7
2	10	89	89.81	91.02	91.085	83.4	83.4	99.4	99.4	90.7	90.7

Table 12 Comparison of proposed models

Models	Overall accuracy (%)
Proposed models:LSTM	99.40
GRU	92.5
Shurman and Khrais [10]	99.19
Cil et al. [12]	94.57
Bolodurina et al. [13]	98.62
Sindian et al. [14]	98

To show the results of the proposed models, the accuracy has been highlighted

dataset is huge. After evaluating the performance of the developed models for each type of attacks, we have also compared their performance against the systems that used the same data set and the results of comparison are presented in Table 12.

Only the models with high accuracy were used for comparison. However, these models are found to have more hidden layers or epochs. The proposed models were able to provide comparable performance with fewer layers or epochs. We assume that fine-tuning hyper-parameters has resulted in improved accuracy. In addition, since we used GPU, the proposed models took less time to practice. Even though, GRU uses time sequence to predict the class, in our attempt, it has given less accuracy compared with other models. Since, the proportion of benign and attack traffic is not balanced in the dataset, the GRU training might not have taken place appropriately. Using sampling techniques like SMOTE, random sampling, etc., the dataset may be balanced and then training can be carried out. This we would like to take up as a future work. The positive side of GRU is that it uses less training parameter and therefore uses less memory and executes faster than LSTM.

6 Conclusion and Future Work

In this work, we investigated the use of deep learning models to create a DDoS attack detection system in this work since deep learning has shown its promise in many domains. We looked at recent research on the topic and discovered that deep learning models can be used to boost the performance of existing systems. To boost accuracy even further, we created RNN-based LSTM and GRU. We evaluated the models in a number of network configurations parameters. These models LSTM and GRU gave an accuracy of 99.4% and 92.5%, respectively, for CICDDoS2019 data set. In the future, we want to expand the models to include very recent data sets with a large number of DDoS attack traces. Other learning models, such as auto encoder and stacked auto encoder, will be tried in the future to increase accuracy and discover new types of DDoS attacks. Also, sampling techniques like SMOTE can be applied on the dataset to address the data imbalance issues. Since the CICDDoS2019 dataset contains more than 80 features and all the features will not contribute for

classification, feature selection algorithms may be employed to reduce the feature set and improve the training accuracy.

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